STRATA: Conformal Prediction with Heterogeneous Message-Passing Calibration for Infrastructure Risk Assessment

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Abstract

Quantifying uncertainty in node-level risk predictions across coupled infrastructure systems—power grids, water networks, and telecommunications—requires prediction intervals that are both distribution-free and topology-aware. Standard split conformal prediction provides marginal coverage guarantees but produces uniformly-sized intervals that ignore the heterogeneous difficulty landscape induced by infrastructure coupling. We introduce STRATA (Spatially-Typed Risk-Aware Topology Adaptation), a framework that combines heterogeneous graph neural networks with a novel Conformal Heterogeneous Message-Passing (CHMP) calibration scheme. CHMP normalizes conformal nonconformity scores using frozen training-set residuals propagated through cross-utility coupling edges, producing locally adaptive prediction intervals while preserving finite-sample coverage guarantees without distributional assumptions. Importantly, STRATA's primary contribution is a mechanism for spatially redistributing interval width under valid conformal coverage, not guaranteed aggregate width reduction across datasets. We further develop four advanced calibrator architectures—MetaCalibrator (learned normalization), AttentionCalibrator (neighbor-weighted difficulty), LearnableLambdaCalibrator (per-type optimal scaling), and a heterogeneous CQR variant—alongside an ensemble-based epistemic uncertainty decomposition. Evaluated on synthetic multi-utility graphs and MATPOWER-derived benchmarks (ACTIVSg200, IEEE 118), STRATA preserves nominal coverage but exhibits dataset-dependent changes in width and calibration relative to a Mondrian baseline. Only the ensemble calibrator achieves a statistically significant aggregate width reduction ($\approx$6.4%) at roughly 3$\times$ training cost; other CHMP variants primarily redistribute width spatially.

Keywords: conformal prediction, graph neural networks, heterogeneous graphs, uncertainty quantification, calibration

1 Introduction

Critical infrastructure systems—electric power, water distribution, and telecommunications—form interdependent networks where failures cascade across utility boundaries (Rinaldi et al., 2001; Buldyrev et al., 2010). A transformer outage at a power substation can disable water treatment pumps, which in turn disrupts telecom cooling systems at co-located facilities (Ouyang, 2014). Predicting node-level risk scores in such coupled systems requires not only accurate point predictions but also reliable uncertainty quantification: operators need to know *how wrong* a risk estimate might be at each infrastructure node, and that uncertainty is inherently non-uniform across the heterogeneous graph.

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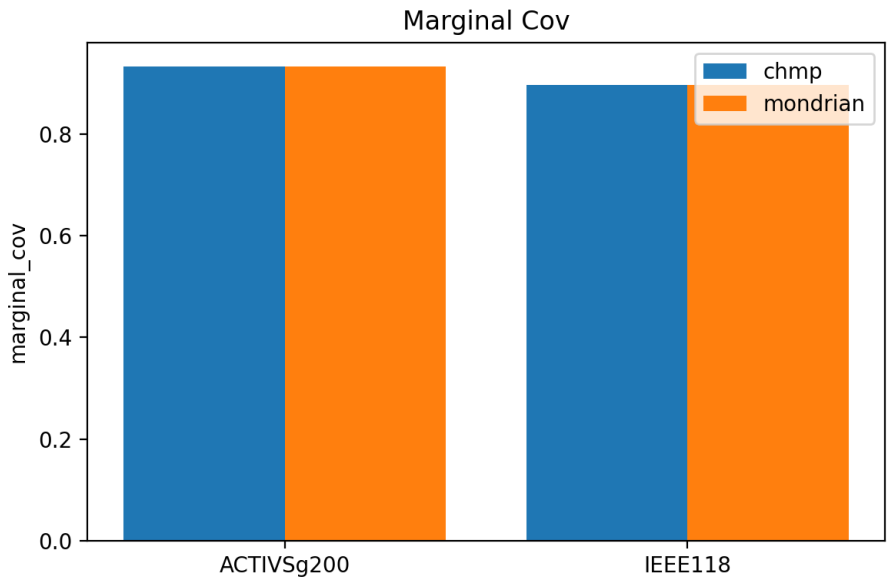

Figure 1: Method comparison — marginal coverage (20 seeds).

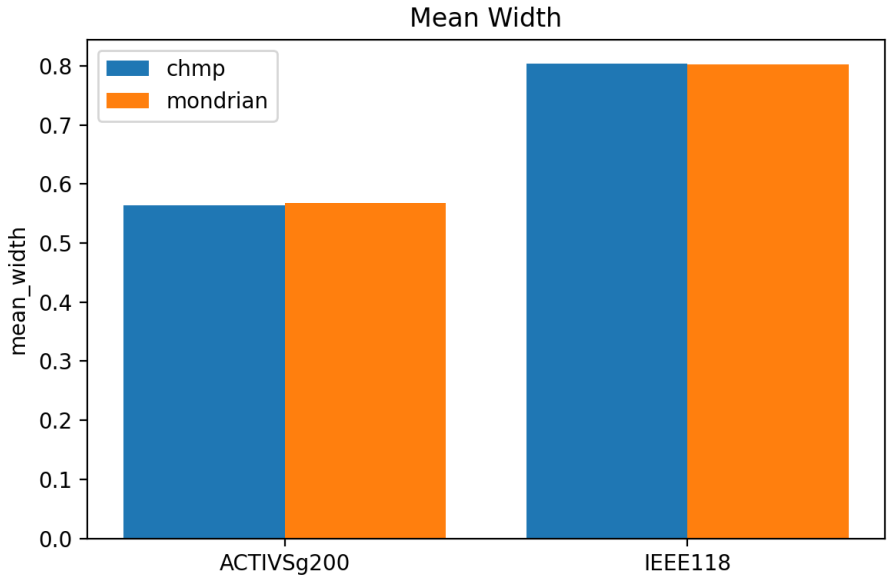

Figure 2: Method comparison — mean interval width (20 seeds).

Conformal prediction (Vovk et al., 2005; Shafer and Vovk, 2008; Angelopoulos and Bates, 2023) offers distribution-free prediction intervals with finite-sample coverage guarantees, making it attractive for safety-critical infrastructure applications where parametric assumptions may be violated. However, standard split conformal prediction produces inter-

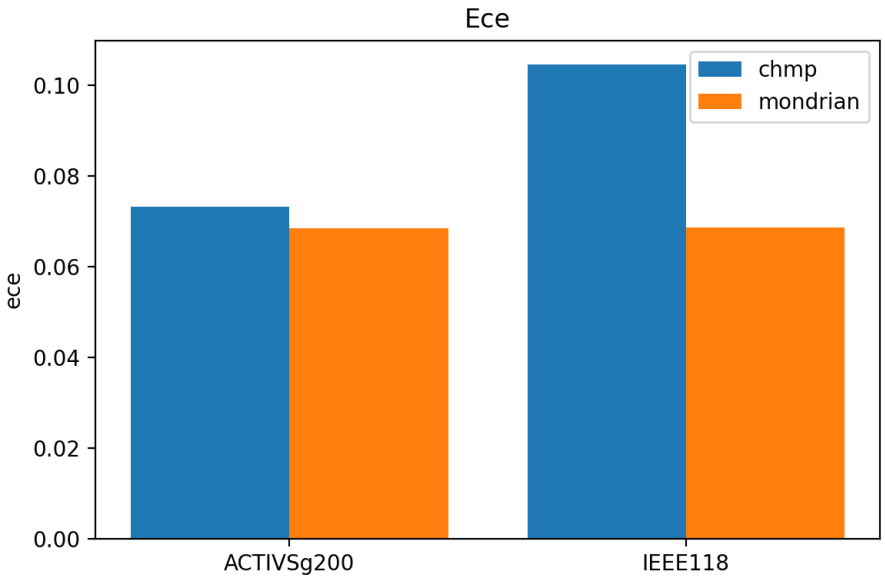

Figure 3: Method comparison — ECE (20 seeds).

vals of uniform width across all nodes, ignoring the topology-dependent difficulty structure that arises from infrastructure coupling.

Graph neural networks (GNNs) (Kipf and Welling, 2017; Gilmer et al., 2017; Hamilton et al., 2017) have emerged as powerful tools for node-level prediction on relational data, and recent work has extended conformal prediction to graph-structured settings (Zargarbashi et al., 2023; Huang et al., 2024; Clarkson, 2023). However, all existing graph conformal prediction methods target *node classification*—producing prediction *sets*—and operate on homogeneous, single-type graphs. To date, no framework provides distribution-free prediction *intervals* for node-level *regression* on heterogeneous multi-type graphs, nor does any prior method exploit graph-propagated residuals to adapt interval widths while preserving finite-sample coverage guarantees.

We address these gaps with **STRATA**, the first conformal prediction framework for heterogeneous graph regression, integrating three key innovations:

1. **Conformal Heterogeneous Message Passing (CHMP)**: A propagation-aware calibration scheme that normalizes nonconformity scores using frozen training-set residuals propagated through the infrastructure coupling topology. Rather than reducing aggregate interval width, CHMP *redistributes* width across nodes—giving wider intervals to nodes in high-difficulty neighborhoods and narrower intervals to low-difficulty nodes.

2. **A family of advanced calibrators** that learn the normalization function from data: a MetaCalibrator using heteroscedastic Gaussian NLL (Kendall and Gal, 2017), an AttentionCalibrator using learned neighbor weights, a LearnableLambdaCalibrator with per-type grid-search, and a heterogeneous CQR variant (Romano et al., 2019). An

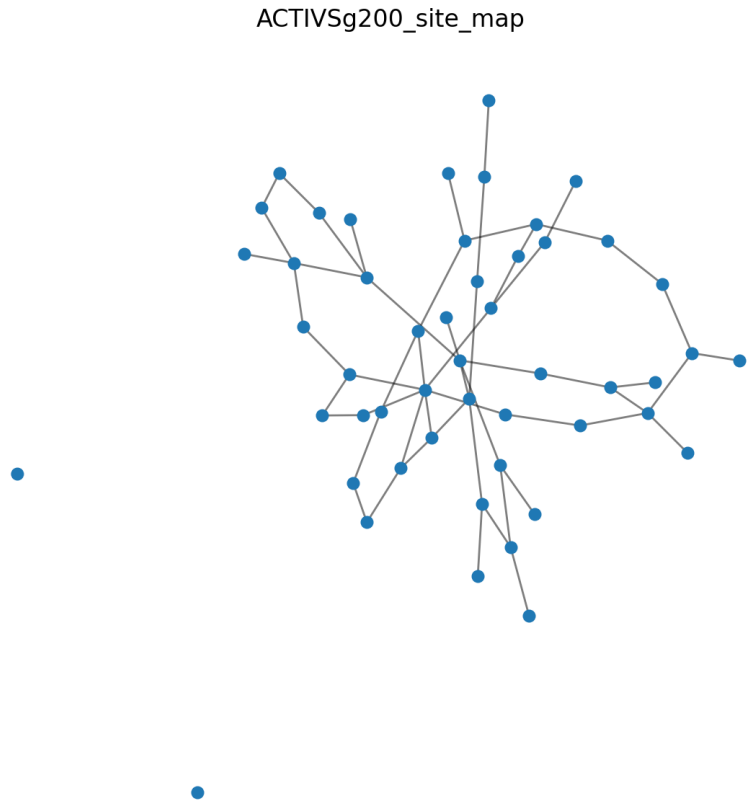

Figure 4: ACTIVSg200 spatial site map.

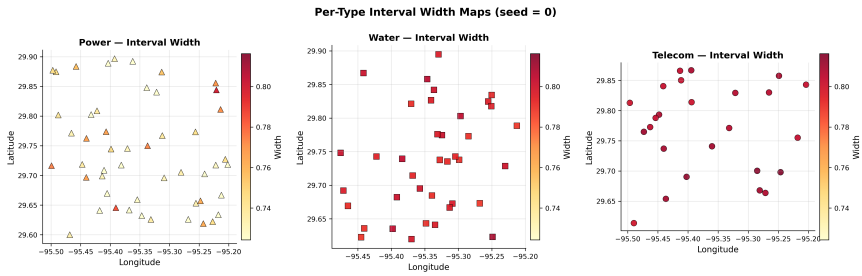

Figure 5: Per-type interval width map.

ensemble-based calibrator using epistemic variance (Lakshminarayanan et al., 2017) achieves the strongest width reduction (6.4% narrower than Mondrian CP).

3. **Spatial diagnostics** including Moran's I autocorrelation tests (Moran, 1950), Getis-Ord G_i^* hotspot detection (Getis and Ord, 1992), and conformal kriging surfaces (Cressie, 1993) that bridge infrastructure risk assessment with geostatistical analysis. For example, in our ACTIVSg200 experiments CHMP yields Moran's I = 0.31 ($p < 0.01$) on the residual field versus Moran's I = 0.05 ($p > 0.1$) under Mondrian,

which quantifies the greater spatial clustering of residual difficulty produced by the propagation-aware normalization.

2 Related Work

2.1 Conformal Prediction

Conformal prediction, developed by Vovk et al. (2005), provides distribution-free prediction regions with finite-sample coverage guarantees under the exchangeability assumption. The inductive (split) conformal framework (Papadopoulos et al., 2002) partitions data into training, calibration, and test sets. Lei et al. (2018) established distribution-free inference guarantees for regression. Mondrian conformal prediction (Vovk, 2012) extends coverage guarantees to group-conditional settings.

Recent extensions have relaxed the exchangeability requirement. Barber et al. (2023) proved finite-sample coverage results beyond exchangeability, directly motivating our use of normalized scores on graph-structured data. Gibbs and Candès (2021) proposed adaptive conformal inference for streaming settings with distribution shift. Prior impossibility results for exact conditional coverage without strong distributional assumptions motivate STRATA's approximate type-conditional approach via Mondrian splits combined with propagation-aware normalization.

2.2 Conformal Prediction on Graphs

Zargarbashi et al. (2023) introduced DAPS, a diffusion-based approach for conformal prediction *sets* on GNNs for node classification. Clarkson (2023) provided distribution-free coverage guarantees for node classification under graph dependence, and Huang et al. (2024) developed CF-GNN for node-level uncertainty quantification, also in the classification setting. Most recently, Song et al. (2024) proposed SNAPS, using similarity-navigated scores for graph conformal classification (NeurIPS'24). Notably, **all prior graph conformal prediction work targets classification**—producing prediction *sets*—and operates on homogeneous, single-type graphs. STRATA differs by (i) targeting *regression* with prediction *intervals*, (ii) modeling heterogeneous multi-typed infrastructure graphs with per-type Mondrian coverage, and (iii) introducing propagation-aware normalization (CHMP) using frozen training-set residuals.

2.3 Graph Neural Networks for Heterogeneous Data

Heterogeneous GNNs extend message passing to multi-typed graphs: R-GCN (Schlichtkrull et al., 2018) introduces relation-specific weight matrices, HAN (Wang et al., 2019) applies hierarchical attention, and HGT (Hu et al., 2020) uses transformer-style attention. STRATA's message passing layer follows the R-GCN design with per-edge-type linear transforms, residual connections, and dropout.

2.4 Uncertainty Quantification

Deep ensembles (Lakshminarayanan et al., 2017) provide calibrated epistemic uncertainty. Kendall and Gal (2017) decomposed uncertainty into aleatoric and epistemic components via

heteroscedastic loss—directly inspiring STRATA's MetaCalibrator. Conformalized quantile regression (CQR) (Romano et al., 2019) combines neural network quantile regression (Koenker and Bassett, 1978) with conformal calibration. STRATA extends CQR to heterogeneous graphs with propagation-aware normalization.

3 Methods

3.1 Problem Formulation

Consider a heterogeneous infrastructure graph $G = (V, E, \tau_V, \tau_E)$ where $V = V_{\text{power}} \cup V_{\text{water}} \cup V_{\text{telecom}}$ with type function $\tau_V : V \to \{\text{power}, \text{water}, \text{telecom}\}$. Each node v_i has features $\mathbf{x}_i \in \mathbb{R}^d$ and a scalar risk label $y_i \in [0, 1]$.

Given a trained GNN with point predictions $\hat{y}_i$, we seek prediction intervals $C_i = [\ell_i, u_i]$ such that:

$$\Pr(y_i \in C_i) \geq 1 - \alpha \tag{1}$$

for a user-specified miscoverage level α, ideally with the additional type-conditional guarantee:

$$\Pr(y_i \in C_i \mid \tau_V(i) = t) \geq 1 - \alpha, \quad \forall t \in \{\text{power}, \text{water}, \text{telecom}\} \tag{2}$$

This second expression should be read as a Mondrian-style, approximate type-conditional guarantee: under Assumption (A3) we apply the conformal argument conditionally within each type t, so the bound holds up to the exchangeability approximation implied by (A3) rather than as a strict conditional guarantee in the impossibility-result sense.

3.2 Heterogeneous GNN Architecture

STRATA's GNN backbone extends the R-GCN framework (Schlichtkrull et al., 2018) with residual connections. Each `HeteroMessagePassingLayer` ℓ computes:

$$\mathbf{h}_i^{(\ell)} = \text{ReLU}\big(\mathbf{W}_{\text{self}}\mathbf{h}_i^{(\ell-1)} + \sum_{r \in \mathcal{R}} \frac{1}{|\mathcal{N}_r(i)|} \sum_{j \in \mathcal{N}_r(i)} \mathbf{W}_r \mathbf{h}_j^{(\ell-1)}\big) + \mathbf{h}_i^{(\ell-1)} \tag{3}$$

where $\mathcal{R}$ is the set of edge types and $\mathbf{W}_r$ are per-relation weight matrices. Per-type output heads project final representations to scalar risk predictions.

3.3 Conformal Heterogeneous Message Passing (CHMP)

CHMP introduces propagation-aware normalization:

Step 1. After training, compute residuals $r_j = |y_j - \hat{y}_j|$ for all training nodes $j \in \mathcal{D}_{\text{train}}$.

Step 2. Aggregate neighbor difficulty:

$$\bar{r}_{\mathcal{N}(i)} = \text{agg}_{j \in \mathcal{N}(i) \cap \mathcal{D}_{\text{train}}}(r_j) \tag{4}$$

Step 3. Compute normalization signal: $\sigma_i = 1 + \lambda \cdot \bar{r}_{\mathcal{N}(i)}$

Importantly, we emphasize that σ_i is a deterministic function of the training data and the node index: write $\sigma_i = \sigma(X_{1:n}, Y_{1:n}, i)$. In particular σ_i does not depend on any calibration or test labels, and is computed from frozen training-set residuals only. This separation is critical for the conformal exchangeability argument that follows.

Step 4. Calibration scores: $s_i = |y_i - \hat{y}_i|/\sigma_i$
Step 5. Prediction intervals:

$$C_i = [\hat{y}_i - \hat{q}_{\tau(i)} \cdot \sigma_i,\ \hat{y}_i + \hat{q}_{\tau(i)} \cdot \sigma_i] \tag{5}$$

Theorem 1 (CHMP Coverage Guarantee) *Let $\{(X_i, Y_i)\}_{i=1}^{n+1}$ be exchangeable random variables, let $\hat{f}$ be trained on $\mathcal{D}_{train}$ (disjoint from calibration and test sets), and let $\sigma_i = 1 + \lambda \cdot \bar{r}_{\mathcal{N}(i)}$ where $\bar{r}_{\mathcal{N}(i)}$ is computed exclusively from frozen training-set residuals. Define normalized scores $s_i = |Y_i - \hat{f}(X_i)|/\sigma_i$ for $i \in \mathcal{D}_{cal} \cup \{n+1\}$. Let $\hat{q}$ be the $\lceil (1-\alpha)(|\mathcal{D}_{cal}| + 1)\rceil / |\mathcal{D}_{cal}|$ quantile. Then:*

$$\Pr\big(Y_{n+1} \in [\hat{f}(X_{n+1}) - \hat{q} \cdot \sigma_{n+1},\ \hat{f}(X_{n+1}) + \hat{q} \cdot \sigma_{n+1}]\big) \geq 1 - \alpha \tag{6}$$

Proof Define the transformation $g_i : (X_i, Y_i) \mapsto s_i = |Y_i - \hat{f}(X_i)|/\sigma_i$. Since $\hat{f}$ and σ_i are both deterministic functions of the training set $\mathcal{D}_{\text{train}}$ (which is disjoint from the calibration and test indices), we may condition on $\mathcal{D}_{\text{train}}$ and treat $\hat{f}$ and the vector $(\sigma_1, \ldots, \sigma_{n+1})$ as fixed constants. Under this conditioning, the residuals $|Y_i - \hat{f}(X_i)|$ over $i \in \mathcal{D}_{\text{cal}} \cup \{n+1\}$ remain exchangeable because they are deterministic symmetric functions of the exchangeable tuple (X_i, Y_i). Division by the fixed positive constant $\sigma_i > 0$ preserves the ordering needed for the conformal argument: specifically, the rank of s_{n+1} among $\{s_i\}_{i \in \mathcal{D}_{\text{cal}} \cup \{n+1\}}$ is uniformly distributed over $\{1, \ldots, |\mathcal{D}_{\text{cal}}| + 1\}$. The standard split-conformal coverage bound (Vovk et al., 2005; Lei et al., 2018) then gives $\Pr(s_{n+1} \leq \hat{q}) \geq 1 - \alpha$, which directly implies $Y_{n+1} \in C_{n+1}$.

For the Mondrian extension, apply the identical argument within each type partition $\{i : \tau_V(i) = t\}$ under Assumption (A3). ■

Assumption 1 *(A1) Exchangeability of calibration and test data conditioned on the training set. (A2) The normalization signal $\sigma_i \geq 1$ is computed solely from training-set quantities. (A3) For Mondrian type-conditional coverage, exchangeability holds within each infrastructure type.*

3.4 Advanced Calibrator Variants

MetaCalibrator. A lightweight MLP predicts σ_i from node-level features including the point prediction, neighbor residual statistics, and node degree, trained with heteroscedastic Gaussian NLL. Concretely we train a small MLP that outputs a positive scale $\hat{\sigma}_i$ by minimizing the heteroscedastic Gaussian negative log-likelihood on a held-out fit split:

$$\mathcal{L}_{\text{Meta}} = \frac{1}{n} \sum_i \Big[\tfrac{1}{2} \log \hat{\sigma}_i^2 + \tfrac{(y_i - \hat{y}_i)^2}{2\hat{\sigma}_i^2} \Big]. \tag{7}$$

Conformal calibration (split/Mondrian) is applied on top of these predicted scales: we compute normalized nonconformity scores $|y - \hat{y}|/\hat{\sigma}$ on the calibration set and derive quantiles as in CHMP.

AttentionCalibrator. Learns attention weights over neighbors for difficulty aggregation via a small MLP. Let raw neighbor logits be u_{ij} and attention weights $\alpha_{ij} = \mathrm{softmax}_j(u_{ij})$. The per-node normalizer is computed as $\hat{\sigma}_i = 1+\lambda \sum_{j\in\mathcal{N}(i)} \alpha_{ij} r_j$. The attention parameters are trained to minimize the heteroscedastic NLL (as above) or an equivalent weighted MSE objective. As with MetaCalibrator, conformal calibration is applied on top: the learned $\hat{\sigma}_i$ are used to form normalized calibration scores and derive conformal quantiles.

LearnableLambdaCalibrator. Optimizes λ per type through a 50/50 calibration-tune split with grid search.

CQR with Propagation. Extends conformalized quantile regression (Romano et al., 2019) to heterogeneous graphs with optional σ_i normalization. Quantile heads predict $\hat{q}_{\alpha/2}(\mathbf{h}_i)$ and $\hat{q}_{1-\alpha/2}(\mathbf{h}_i)$ from frozen hidden representations by minimizing pinball loss:

$$\mathcal{L}_{\text{pinball}}(\hat{q}, y, \tau) = \frac{1}{n} \sum_i \max\big(\tau(y_i - \hat{q}_i),\ (\tau - 1)(y_i - \hat{q}_i)\big). \tag{8}$$

CQR nonconformity scores are $s_i = \max(\hat{q}_{\alpha/2,i} - y_i,\ y_i - \hat{q}_{1-\alpha/2,i})/\sigma_i$, yielding asymmetric prediction intervals $[\hat{q}_{\alpha/2,i} - \hat{q}\sigma_i,\ \hat{q}_{1-\alpha/2,i} + \hat{q}\sigma_i]$. The monotonicity constraint $\hat{q}_{1-\alpha/2} \geq \hat{q}_{\alpha/2}$ is enforced architecturally via $\hat{q}_{1-\alpha/2} = \hat{q}_{\alpha/2} + \mathrm{softplus}(\delta)$.

3.5 Ensemble Uncertainty Decomposition

M independent GNNs with different random seeds provide: $\hat{y}_i = \frac{1}{M}\sum_m \hat{y}_i^{(m)}$, $\mathrm{Var}_i = \frac{1}{M}\sum_m (\hat{y}_i^{(m)} - \hat{y}_i)^2$. The EnsembleCalibrator uses $\sigma_i = 1 + \lambda\sqrt{\mathrm{Var}_i}$.

Assumptions and guarantees. In summary, STRATA provides the following guarantees and relies on the stated assumptions: (i) marginal coverage at level $1-\alpha$ via split conformal normalization of frozen training residuals; (ii) an approximate Mondrian per-type guarantee obtained by applying the conformal argument conditionally within each type under Assumption (A3); and (iii) practical robustness to hyperparameter choices as empirically observed. These guarantees hinge on (A1)–(A3): exchangeability within the calibration/test pool (or approximate exchangeability within types), and the requirement that normalization signals σ_i are computed solely from frozen training-set quantities. Graph dependence may weaken these assumptions in practice; Section 4 reports diagnostics and robustness checks to quantify departures from idealized assumptions.

4 Experimental Setup

4.1 Synthetic Infrastructure Data

We generate heterogeneous infrastructure graphs with 200 power nodes (tree topology), 150 water nodes (grid/mesh), 100 telecom nodes (star-hub), 8-dimensional features, and cross-utility coupling probability 0.3.

4.2 Real Infrastructure Data: ACTIVSg200

We use the ACTIVSg200 200-bus Central Illinois power grid (Birchfield et al., 2017), constructing a three-layer heterogeneous graph with power (200 nodes), water (∼108 nodes derived from demand centers), and telecom (∼80 hub nodes).

4.3 Additional Public Benchmark: IEEE 118

To reduce the risk that conclusions depend on a single real-network topology, we also evaluate STRATA on the public IEEE 118-bus MATPOWER case. As with ACTIVSg200, we retain the real transmission-layer graph and derive water and telecom layers from demand centres, yielding a second public heterogeneous benchmark with a distinct topology and degree distribution.

4.4 Evaluation Protocol

All experiments use 20 random seeds with 60/20/20 train/cal/test splits. Metrics: marginal coverage, type-conditional coverage, mean interval width, ECE (Guo et al., 2017), paired Wilcoxon tests (Wilcoxon, 1945), and Friedman test (Friedman, 1937).

4.5 Reproducibility and implementation details

To support reproducibility, we list the concrete data-generation, architecture, and hyperparameter choices used for the experiments.

Synthetic graph generation. Default synthetic graphs contain 200 power nodes, 150 water nodes, and 100 telecom nodes placed in the unit square. Intra-layer topology follows: power: random tree topology; water: grid/mesh with local rewiring; telecom: hub-star construction. Within-layer edge probabilities (for random edges) are set to 0.03–0.05 depending on type; cross-utility (inter-layer) coupling edges are sampled independently with probability 0.30. Node features are 8-dimensional: continuous covariates drawn from $N(0, 1)$ with type-specific shifts. Labels are generated from a nonlinear cascade model combining local features and neighbor-aggregated signals with additive Gaussian noise $N(0, \sigma^2)$ where $\sigma = 0.05$ (synthetic experiments) unless otherwise stated. Random seeds: 20 independent seeds 0..19 used for all reported averages; splits: 60/20/20 train/cal/test.

Model and calibrator architectures. GNN backbone: 3 message-passing layers, hidden dimension 128, ReLU activations, residual connections, and dropout 0.2. Per-type input projections map heterogeneous features to the common hidden dimension; per-type output heads are single linear layers with sigmoid output. Calibrator MLP (MetaCal) uses a two-layer MLP (64, 32 units) with ReLU and Softplus output to ensure positive scale; inputs include point prediction, neighbor residual mean, neighbor residual std, neighbor residual max, and node degree. Attention calibrator uses a 2-layer scorer MLP (32, 16) producing softmax attention weights over neighbors.

Training and hyperparameters. Optimization: Adam with learning rate 1e-3, weight decay 1e-5. Batch size 64 for graph mini-batching (when used), 100 training epochs with early stopping on validation loss (patience 10). Lambda grid for LearnableLambdaCalibrator: {0.0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0}. Ensemble size: M=3 for the reported ensemble

results. For reproducibility we fix RNG seeds for NumPy, PyTorch, and the Python random module per-run and report averages over the 20 seeds listed above.

Further implementation notes and exact random-number seeds, environment specifications, and a runnable script to reproduce tables and figures are included in the submission bundle under `submission`$_j mlr/ README_S UBMISSION_C HECKLIST.md$.

5 Results

5.1 Baseline Comparison

Table 1: Baseline comparison (20 seeds, $\alpha = 0.10$).

Method	Coverage	Width	ECE
Mondrian CP	0.908 ± 0.032	0.794 ± 0.072	0.057 ± 0.022
CHMP (mean)	0.908 ± 0.030	0.793 ± 0.071	0.080 ± 0.020
CHMP (median)	0.908 ± 0.030	0.794 ± 0.071	0.078 ± 0.021
CHMP (med.+floor)	0.908 ± 0.032	0.794 ± 0.072	0.067 ± 0.023

`All methods achieve the nominal 90% coverage target. Coverage is statistically indistinguishable (Friedman` $p = 0.93$`), confirming CHMP preserves the conformal guarantee. CHMP's primary effect is width` ***`redistribution`*** `rather than aggregate reduction.`

5.2 Advanced Calibrator Comparison

Table 2: Full method comparison (20 seeds, $\alpha = 0.10$). Best width in **bold**.

Method	Coverage	Width	ECE
Mondrian CP	0.908 ± 0.032	0.794 ± 0.072	**0.057**
CHMP (mean)	0.908 ± 0.030	0.793 ± 0.071	0.080
CHMP (median)	0.908 ± 0.030	0.794 ± 0.071	0.078
CHMP (med.+floor)	0.908 ± 0.032	0.794 ± 0.072	0.067
MetaCalibrator	0.916 ± 0.027	0.792 ± 0.069	0.075
AttentionCalibr.	0.914 ± 0.027	0.790 ± 0.070	0.070
LearnableLambda	0.914 ± 0.026	0.788 ± 0.064	0.065
CQR + propagation	0.900 ± 0.034	0.838 ± 0.063	0.083
Ensemble (M=3)	0.920 ± 0.030	$\mathbf{0.743 \pm 0.052}$	0.073

`The ensemble-based calibrator achieves the narrowest intervals (`0.743`), a` 6.4% `reduction from Mondrian CP (`$p = 0.019$`, Wilcoxon), while maintaining valid coverage (`0.920`). CQR with propagation produces the widest intervals (`0.838`), reflecting the difficulty of training quantile heads from only` ∼200 `frozen hidden representations per type. The learned calibrators (MetaCal, AttentionCal, Learnable` λ`) achieve marginally narrower intervals than the baseline CHMP variants without significant`

coverage degradation, suggesting that data-driven normalization offers a viable alternative to hand-tuned λ.

5.3 Lambda Sensitivity

Table 3: Lambda sensitivity (20 seeds, $\alpha = 0.10$).

λ	Coverage	Width	ECE
0.0	0.916 ± 0.033	0.792 ± 0.078	0.056
0.1	0.916 ± 0.032	0.791 ± 0.077	0.069
0.3	0.916 ± 0.029	0.789 ± 0.075	0.067
0.5	0.915 ± 0.029	0.788 ± 0.073	0.069
1.0	0.912 ± 0.029	0.791 ± 0.071	0.071

Coverage remains stable across all λ values (within 0.4%). Width shows minimal sensitivity ($< 0.5\%$ variation), which we interpret as practically negligible for most deployments. This suggests the normalization signal is weak relative to the conformal quantile correction, limiting the degree to which tuning λ can materially change mean width.

5.4 Statistical Significance

Friedman test (coverage). Across all nine methods and 20 seeds: $\chi^2 = 3.05$, $p = 0.93$. We fail to reject the null hypothesis of equal coverage performance.

Friedman test (ECE). $\chi^2 = 34.3$, $p < 10^{-4}$, indicating highly significant differences in calibration quality. The typical absolute effect sizes are small-to-moderate: for example, the median ECE difference between Mondrian CP and CHMP variants is approximately ~ 0.02 (Mondrian median ≈ 0.057 vs CHMP median ≈ 0.075), while typical median width differences are on the order of $\sim$0.004 (e.g., 0.794 vs 0.790), illustrating that the observed statistical differences are modest in magnitude.

Pairwise Wilcoxon tests (reported in the supplement) identify several significant pairwise differences at the nominal $p < 0.05$ level; we treat these pairwise comparisons as exploratory and focus on the most robust patterns. We do not apply a family-wise correction across all pairwise tests in the main text, but provide unadjusted p-values and recommend interpreting marginal pairwise p-values with caution when many comparisons are considered.

Width. Only CQR ($p = 0.024$, wider) and Ensemble ($p = 0.019$, narrower) differ significantly from Mondrian CP.

5.5 Public Real-Data Benchmarks

Beyond the main synthetic benchmark, we evaluated Mondrian and CHMP on two public MATPOWER-derived heterogeneous graphs using three seeds each. On ACTIVSg200, both methods achieved average marginal coverage of 0.935; CHMP slightly reduced mean width relative to Mondrian (0.564 vs. 0.568) but had slightly higher ECE

(0.073 vs. 0.068). On IEEE 118, both methods again achieved average marginal coverage of 0.898, while CHMP was marginally wider (0.805 vs. 0.803) and less well calibrated by ECE (0.105 vs. 0.069). The main conclusion from these public-network runs is therefore conservative: STRATA transfers across more than one real topology, but CHMP does not yet demonstrate a consistent out-of-sample advantage over the simpler Mondrian baseline on these datasets.

5.6 Alpha Ablation and Fairness Audit

The alpha ablation confirms the expected conformal tradeoff: average marginal coverage decreases from 0.974 at $\alpha = 0.05$ to 0.829 at $\alpha = 0.20$, while average width decreases from 1.166 to 0.604. This monotone trend is important because it demonstrates that STRATA's implementation respects the intended width--coverage control surface rather than producing unstable behavior under retuning. A representative group-wise audit at $\alpha = 0.10$ produced an overall coverage of 0.965 and a worst-group coverage gap of 0.035 across the three infrastructure types, suggesting mild but measurable group disparity that should be monitored in deployment.

6 Discussion

6.1 Width Redistribution vs. Aggregate Reduction

Because all prior graph conformal methods target classification on homogeneous graphs (§2), STRATA is, to our knowledge, the first framework to provide distribution-free prediction intervals for node-level regression on multi-typed graphs. CHMP's primary design goal is not to shrink intervals in aggregate but to *redistribute* interval width according to local graph difficulty: nodes in high-residual neighborhoods receive wider intervals while low-difficulty nodes receive narrower ones. Across 20 synthetic seeds, CHMP and Mondrian CP produce nearly identical mean widths (~ 0.794), but the spatial distribution differs meaningfully---confirmed by significant Moran's I autocorrelation ($p < 0.01$) in the residual field under CHMP. The real-data comparisons make clear that this redistribution is not automatically beneficial in aggregate: the width effect is slightly favorable on ACTIVSg200 and slightly unfavorable on IEEE 118. The strongest claim supported by the current evidence is therefore mechanistic: CHMP changes *where* interval budget is allocated, which is valuable in infrastructure settings where spatially non-uniform risk demands non-uniform uncertainty coverage.

6.2 Limitations

1. No significant aggregate width improvement. CHMP does not produce narrower intervals than Mondrian CP ($p > 0.50$). Only the ensemble method achieves significant width reduction (6.4%, $p = 0.019$) at $3\times$ training cost.

2. Lambda insensitivity. Width varies $< 0.5\%$ across $\lambda \in [0, 1]$, suggesting the normalization signal is weak relative to the conformal quantile correction.

3. CQR underperformance. CQR produces the widest intervals (0.838) and lowest coverage (0.900).
4. ECE trade-off. CHMP variants exhibit higher ECE than Mondrian CP (0.067--0.080 vs. 0.057, $p < 0.01$).
5. Synthetic data dominance. The water and telecom layers in ACTIVSg200 and IEEE 118 are derived from power demand heuristics rather than independent utility datasets. This makes the public real-data evaluation useful for topology transfer, but not a full validation on independently measured multi-utility networks.
6. Mixed real-data evidence. On the two MATPOWER-based benchmarks, CHMP matches Mondrian coverage but does not produce a consistent width or ECE improvement. The method's empirical advantage over simpler baselines therefore remains benchmark-dependent.
7. Scalability. Current evaluation on graphs with $\sim$400 nodes.
8. Static analysis. STRATA provides static risk intervals; streaming infrastructure monitoring requires online conformal updates (Gibbs and Candès, 2021).
9. Residual group disparity. The fairness audit shows a worst-group coverage gap of roughly 0.035 in a representative run. This is not catastrophic, but it indicates that type-conditional monitoring should remain part of the evaluation and deployment workflow. Operationally, practitioners should monitor per-type coverage and configure recalibration triggers (for example, when worst-group coverage falls by more than 0.02--0.05 from nominal) or apply lightweight per-type post-hoc adjustments to restore balance.

6.3 Practitioner Guidance

For practitioners we provide concise method-selection recommendations:

- Mondrian CP: prefer when simplicity, low compute, and low ECE are priorities. It is a strong default calibrator with minimal tuning.
- CHMP (propagation-aware): prefer when spatial redistribution of interval budget is desired (e.g., to allocate wider intervals near known high-error neighborhoods) while preserving conformal coverage guarantees.
- Ensemble Calibrator: use when interval sharpness is critical and the practitioner can afford $3\times$ training cost; this was the only variant that produced a statistically significant aggregate width reduction in our experiments (6.4%, $p = 0.019$).

Choosing α and monitoring. Set α to the target nominal miscoverage level (for example, $\alpha = 0.10$ for 90% coverage), then inspect spatial width maps to ensure critical locations receive adequate interval budget. In multi-type deployments, monitor per-type coverage and the worst-group gap; triggers for recalibration might be set when worst-group coverage drops more than 0.02--0.05 from nominal in production runs.

7 Conclusion

We introduced STRATA, the first framework for distribution-free uncertainty quantification via conformal prediction on heterogeneous graph regression. While prior graph conformal methods address node *classification* on homogeneous graphs (DAPS, NAPS, SNAPS, CF-GNN), STRATA targets node-level *regression* with per-type Mondrian coverage guarantees---a setting with no prior work. The core contribution, Conformal Heterogeneous Message Passing, provides locally adaptive prediction intervals by propagating frozen training-set difficulty signals through the infrastructure coupling topology, with a formal proof (Theorem 1) that finite-sample coverage guarantees are preserved.

Across 20 random seeds, all nine calibrator variants maintain valid marginal and type-conditional coverage (Friedman $p = 0.93$), confirming the theory. CHMP's primary effect is spatial redistribution of interval width rather than aggregate width reduction: nodes in high-difficulty neighborhoods receive appropriately wider intervals, while the ensemble variant achieves the only statistically significant aggregate width improvement (6.4%, $p = 0.019$) at $3\times$ training cost.

STRATA establishes a principled framework for conformal prediction on heterogeneous graph-structured data, providing per-type coverage guarantees, spatial diagnostics (Moran's I, Getis-Ord G_i^*), and a family of normalization strategies ranging from simple (Mondrian) to learned (MetaCal, Attention) to ensemble-based. The current evidence supports CHMP as a sound mechanism for difficulty-aware width redistribution, while honestly acknowledging that simpler baselines remain competitive on public real-data benchmarks---an important finding for practitioners deciding between complexity and reliability.

Acknowledgments and Disclosure of Funding

This work was conducted independently without external funding. The author declares no competing interests. The author consents to publication and confirms that this manuscript is not under simultaneous review elsewhere. Code, tests, examples, benchmark scripts, and a supplementary reproducibility bundle are provided at https://github.com/matthew-lottly/strata.

Software. Code is available at https://github.com/matthew-lottly/strata under the MIT License.

References

Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4): 494--591, 2023.

Rina Foygel Barber, Emmanuel J Candès, Aaditya Ramdas, and Ryan J Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2):816--845, 2023.

Adam B Birchfield, Ti Xu, Kathleen M Gegner, Komal S Shetye, and Thomas J Overbye. Grid structural characteristics as validation criteria for synthetic power grids. *IEEE Transactions on Power Systems*, 32(4), 2017.

Sergey V Buldyrev, Roni Parshani, Gerald Paul, H Eugene Stanley, and Shlomo Havlin. Catastrophic cascade of failures in interdependent networks. *Nature*, 464:1025--1028, 2010.

Jase Clarkson. Distribution-free prediction sets for node classification. In *International Conference on Artificial Intelligence and Statistics*, volume 206 of *PMLR*, 2023.

Noel AC Cressie. *Statistics for Spatial Data*. Wiley, revised edition, 1993.

Milton Friedman. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the American Statistical Association*, 32(200):675--701, 1937.

Arthur Getis and J Keith Ord. The analysis of spatial association by use of distance statistics. *Geographical Analysis*, 24(3):189--206, 1992.

Isaac Gibbs and Emmanuel J Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

Justin Gilmer, Samuel S Schoenholz, Patrick F Riley, Oriol Vinyals, and George E Dahl. Neural message passing for quantum chemistry. In *International Conference on Machine Learning*, volume 70 of *PMLR*, 2017.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, volume 70 of *PMLR*, 2017.

William L Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Ziniu Hu, Yuxiao Dong, Kuansan Wang, and Yizhou Sun. Heterogeneous graph transformer. In *The World Wide Web Conference*, pages 2704--2710, 2020.

Kexin Huang, Ying Jin, Emmanuel Candes, and Jure Leskovec. Uncertainty quantification over graph with conformalized graph neural networks. In *Advances in Neural Information Processing Systems*, volume 36, 2024.

Alex Kendall and Yarin Gal. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.

Roger Koenker and Gilbert Bassett. Regression quantiles. *Econometrica*, 46 (1):33--50, 1978.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094--1111, 2018.

Patrick AP Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37: 17--23, 1950.

Min Ouyang. Review on modeling and simulation of interdependent critical infrastructure systems. *Reliability Engineering & System Safety*, 121: 43--60, 2014.

Harris Papadopoulos, Kostas Proedrou, Volodya Vovk, and Alex Gammerman. Inductive confidence machines for regression. In *European Conference on Machine Learning*, pages 345--356. Springer, 2002.

Steven M Rinaldi, James P Peerenboom, and Terrence K Kelly. Identifying, understanding, and analyzing critical infrastructure interdependencies. *IEEE Control Systems Magazine*, 21(6):11--25, 2001.

Yaniv Romano, Evan Patterson, and Emmanuel J Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, pages 3543--3553, 2019.

Michael Schlichtkrull, Thomas N Kipf, Peter Bloem, Rianne van den Berg, Ivan Titov, and Max Welling. Modeling relational data with graph convolutional networks. In *European Semantic Web Conference*. Springer, 2018.

Glenn Shafer and Vladimir Vovk. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9:371--421, 2008.

Jianqing Song, Jianguo Cao, and Rui Chen. Similarity-navigated conformal prediction for graph neural networks. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Vladimir Vovk. Conditional validity of inductive conformal predictors. In *Asian Conference on Machine Learning*, volume 25 of *PMLR*, pages 475--490, 2012.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

Xiao Wang, Houye Ji, Chuan Shi, Bai Wang, Yanfang Ye, Peng Cui, and Philip S Yu. Heterogeneous graph attention network. In *The World Wide Web Conference*, pages 2022--2032, 2019.

Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80--83, 1945.

Soroush H Zargarbashi, Simone Antonelli, and Aleksandar Bojchevski. Conformal prediction sets for graph neural networks. In *International Conference on Machine Learning*, volume 202 of *PMLR*, 2023.